CRMLint: Automated Logical Cost Analysis of Mathematical Proof

Paper 99, of the Constructive Reverse Mathematics Series

A Methodological Paper in Six Parts

Paul Chun-Kit Lee
Division of Cardiology, NYU Grossman School of Medicine
paul.lee@nyulangone.org

March 2026

Abstract

We describe the design, calibration, and first external deployment of CRMLint, a heuristic tool for estimating the constructive reverse mathematics (CRM) signature of mathematical proofs. CRMLint operates in three modes: as a Lean 4 proof-term analyzer, as a structural proof profiler, and as a lightweight LaTeX scanner that classifies papers by keyword and citation pattern matching.

Calibration against 627 declarations across 10 Mathlib4 files recovers the expected algebraic/analytic divide: 92.0% of declarations are BISH, with classical content concentrating in inner product spaces and convergence arguments. A three-axis structural analysis finds that Archimedean proofs are 1.4× more complex, 2.8× deeper in dependency chains, and carry 4.3× more external lemma invocations than their non-Archimedean counterparts.

We deploy the LaTeX scanner against the 15-file blueprint of the Liquid Tensor Experiment (LTE), Scholze's formally verified proof of Theorem 9.4 on Ext-vanishing for p'-liquid real vector spaces. The initial scan returns 86.7% BISH—a gross undercount caused by the scanner's inability to detect norm-comparison content that is structurally present but does not match any keyword pattern. We call this *Archimedean dark matter*. Patching the dictionary to detect norm-filtration patterns revises the profile to 33.3% BISH / 66.7% LPO.

We then annotate the LTE's mathematical content—the main theorem, its six-link proof chain, and the normed homological algebra toolkit—with CRM levels at each step. This is exposition, not discovery: the concentration of classical content at norm-filtration gates is a design feature of condensed mathematics, understood by its creators. Our contribution is a vocabulary (the CRM hierarchy provides a graduated scale where before there was a binary algebra/analysis distinction) and a tool (the scanner automates detection of this structure for proofs where the architecture is not designed).

The paper is honest about what it does not contribute: no new mathematics, no simplification of the LTE, no result that would surprise Scholze.

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Part I

Genesis

1 The problem

After 96 papers of manual CRM audit—classifying theorems one at a time, tracing axiom dependencies by hand, flagging each use of excluded middle or Archimedean completeness—the bottleneck in the program shifted from mathematics to labor. The CRM classification of a single theorem (say, the Gross–Zagier formula) requires reading the proof, identifying every appeal to a non-constructive principle, and tracing each appeal to its minimal logical strength. Papers 78–96 each took 2–5 days of combined human and AI effort. Scaling this to the full Mathlib library (∼200,000 declarations as of early 2026) is infeasible by hand.

The question that opens this paper is operational: *can the CRM classification of a proof be automated?*

2 What CRM measures

CRM assigns to each proof a *logical signature* $\sigma(P)$, the weakest axiom system in which the proof can be carried out. The hierarchy is:

$$\text{BISH} \subset \text{BISH+MP} \subset \text{BISH+LLPO} \subset \text{BISH+WKL} \subset \text{BISH+WLPO} \subset \text{BISH+LPO} \subset \text{CLASS}$$

A theorem with signature BISH requires no omniscience principle whatsoever: every step is computationally witnessed. A theorem with signature CLASS invokes the full law of excluded middle somewhere in its proof tree. The intermediate levels—MP, LLPO, WKL, WLPO, LPO—measure calibrated degrees of non-constructivity, each corresponding to a specific decision problem on infinite sequences or real numbers.

The CRM signature is *not* a measure of proof complexity. Taylor–Wiles patching is technically intricate but logically WKL (an inverse limit of finite objects). The intermediate value theorem is elementary but logically WLPO (it requires deciding whether a real number equals zero). CRM measures the *logical cost* of a proof—the price paid for treating the continuum as a completed object with decidable properties.

3 Three generations of automation

CRMLint was developed in three stages, each addressing a different aspect of the automation problem:

Generation 1: Theoretical framework (`crmlint.py`, 1,161 lines). Models the CRM hierarchy, six realization functors (Betti, de Rham, étale, crystalline, Hodge, automorphic) with their CRM costs, comparison isomorphisms, and the Archimedean Obstruction Theorem of Paper 98. Proof DAGs compute signatures by max-propagation over dependency edges. A substitution database stores eight seed entries extracted from Papers 50–97: known techniques where a classical method can be replaced by a constructive alternative at lower logical cost.

This generation answers: *what is the right mathematical model for CRM signatures of composite proofs?*

Generation 2: Empirical validation (`crmlint_crawler.py`, 653 lines; `crmlint_structure.py`, 793 lines). The crawler fetches Lean 4 source files from Mathlib4 on GitHub, parses declara-

tions, and heuristic-classifies each by CRM level using tactic and keyword signatures (∼40 classical indicators including `by_contra`, `Classical.choice`, inner product space infrastructure, integration, and topological convergence patterns). The structure analyzer computes seven-dimensional proof profiles: proof size (lines, tactic count), dependency depth (`have`/`let` chains), proof width (active hypotheses, case splits), and modularity (automation ratio, external lemma invocations, rewrite density).

This generation answers: *does the CRM hierarchy correlate with structural proof complexity in real Lean code?*

Generation 3: LaTeX scanner (`crmlint_scanner.py`, 810 lines at v1.0, 870 lines at v1.1). Keyword and citation pattern matching against ∼90 technique markers and ∼120 landmark paper citations, applied directly to `.tex` source. No Lean, no LLM, no external dependencies beyond the Python standard library. Scans a directory of papers in seconds. Anti-marker patterns suppress false positives (e.g., algebraic "integral closure" is not analytic integration).

This generation answers: *can we triage thousands of papers by CRM cost before investing months in any one?*

The three generations are complementary. Generation 1 provides the theory; Generation 2 tests it against ground truth; Generation 3 scales it to the full mathematical literature. All three are implemented in pure Python with zero external dependencies (3,417 lines total at v1.0).

Part II

Calibration: Mathlib at 627 Declarations

4 Target selection

We selected 10 Mathlib4 files spanning eight mathematical domains:

File	Domain	Decls
`EllipticCurve/Weierstrass.lean`	Algebraic Geometry	68
`InnerProductSpace/Basic.lean`	Analysis	141
`SpecificLimits/Basic.lean`	Analysis	83
`Galois/Basic.lean`	Field Theory	61
`Charpoly/Basic.lean`	Linear Algebra	39
`ArithmeticFunction.lean`	Number Theory	0
`Bernoulli.lean`	Number Theory	27
`RepresentationTheory/Basic.lean`	Representation Theory	67
`Cyclotomic/Basic.lean`	Ring Theory	57
`InfiniteSum/Basic.lean`	Topology	84

The selection was guided by two criteria: (1) coverage of the algebraic/analytic divide that the Archimedean Obstruction Thesis predicts to be the primary CRM boundary, and (2) inclusion of domains where the CRM series has ground-truth classifications from manual audit (inner product spaces from Paper 77, elliptic curves from Paper 95, Galois theory from Paper 78).

5 Generation 2 results: the crawler

The crawler classified 627 declarations:

CRM Level	Count	%
BISH	577	92.0
WKL	11	1.8
WLPO	20	3.2
CLASS	19	3.0

The 92% BISH figure is consistent with the programme's central finding: most of mathematics, as actually practiced in a modern proof library, is constructive. The 8% non-BISH content concentrates in three locations:

(i) **Inner product spaces** (Analysis): 6 declarations at CLASS, 4 at WLPO. The inner product on a Hilbert space is an inherently Archimedean object: $\langle x, x\rangle \geq 0$ with equality iff $x = 0$ requires decidable equality of reals. The Cauchy–Schwarz inequality $|\langle x, y\rangle| \leq \|x\|\|y\|$ requires comparison of real norms. This is exactly the Archimedean obstruction of Papers 5 and 77.

(ii) **Specific limits** (Analysis): 1 declaration at CLASS, 11 at WKL. The geometric series $\sum_{n=0}^{\infty} r^n = 1/(1-r)$ for $|r| < 1$ involves convergence of a real sequence—decidability of whether the partial sums are within ϵ of the limit. The single CLASS declaration (`tendsto_pow_atTop_nhds_zero_of_lt_one`) uses `by_contra`, a potentially eliminable appeal to excluded middle.

(iii) **Galois theory** (Field Theory): 12 declarations at CLASS. A false positive cluster: the crawler flags uses of "integral" (meaning integral closure, an algebraic concept) as analytic integration. This motivated the anti-marker system in the Generation 3 scanner.

6 Per-domain cross-tabulation

Domain	N	BISH	WKL	WLPO	CLASS
Linear Algebra	39	39	0	0	0
Ring Theory	57	57	0	0	0
Number Theory	27	27	0	0	0
Topology	84	84	0	0	0
Algebraic Geometry	68	66	0	2	0
Representation Th.	67	55	0	12	0
Analysis	224	202	11	4	7
Field Theory	61	47	0	2	12

The algebraic/analytic divide is sharp. Four purely algebraic domains (linear algebra, ring theory, number theory, topology) are 100% BISH across 207 declarations. The two mixed domains (algebraic geometry, representation theory) show WLPO leakage from `noncomputable` definitions—the Lean 4 artifact of working over fields where equality is not decidable—but no CLASS content. The two analytic domains (analysis, field theory) contain all 19 CLASS declarations.

The field theory anomaly (algebraic domain, CLASS content) is instructive. Manual inspection reveals that the 12 flagged declarations use the word "integral" in the sense of integral ring extensions, not integration. The crawler's keyword dictionary conflated the two senses. This

false positive directly motivated the anti-marker system described in §17.

Part III

The Archimedean Obstruction, Quantified

7 The three-axis profiler

Generation 2 includes a structural profiler (`crmlint_structure.py`) that measures each proof along three axes orthogonal to CRM level:

Axis 1: Size. Lines of proof, tactic invocations, term size.

Axis 2: Depth. Length of the longest `have`/`let` chain; maximum nesting of dependent intermediate results.

Axis 3: Width. Number of simultaneously active hypotheses; cyclomatic complexity (number of independent case branches).

An additional modularity metric records the automation ratio (fraction of proof steps discharged by `simp`, `ring`, or `norm_num` without human guidance) and the external lemma count (number of `exact`/`apply` calls referencing results outside the current file).

8 Aggregate statistics

Over 627 declarations:

Metric	Mean	Median	Max
Proof lines	5.0	4	57
Dependency depth	0.3	0	8
Active hypotheses	0.4	0	8
Cyclomatic complexity	1.5	1	15
Overall complexity	0.038	0.026	0.479

The median proof is 4 lines with no intermediate dependencies and no case splits. The distribution is strongly right-skewed: 70.7% of proofs are classified as `TRIVIAL` (one-step `simp`, `rfl`, or direct `exact`).

9 Tactic frequency

Tactic	Count	%
`simp`	577	27.5
`rw`	448	21.3
`exact`	214	10.2
`have`	167	7.9
`rfl`	82	3.9
`ext`	80	3.8
`refine`	80	3.8
`apply`	79	3.8
`intro`	72	3.4

Three tactics (`simp`, `rw`, `exact`) account for 59% of all tactic invocations. The high `simp` frequency reflects MATHLIB4's design philosophy: push complexity into the simp lemma database, keep proofs short. Crucially, `simp` is BISH-safe: it applies equational rewriting, never introduces classical content.

10 The Archimedean structural gap

We partitioned the 627 declarations into those touching Archimedean content (43 declarations: inner product, real convergence, norm comparison) and the remainder (584 declarations).

Metric	Archimedean	Non-Arch.	Ratio
Avg. complexity score	0.05	0.04	1.4×
Avg. proof lines	7.95	4.75	1.7×
Avg. dependency depth	0.72	0.26	2.8×
Avg. active hypotheses	0.81	0.37	2.2×
Avg. external lemmas	0.65	0.15	4.3×
Avg. automation ratio	0.22	0.39	0.56×

Every axis tells the same story. Archimedean proofs are longer, deeper, wider, more externally coupled, and less automatable than non-Archimedean proofs. The automation ratio drops from 39% to 22%: when real-number arguments enter, `simp` and `ring` give way to `have` chains, explicit ϵ-δ management, and manual case analysis.

This is the Archimedean Obstruction Thesis of Papers 5–77 made quantitative: *the logical cost of $\mathbb{R}$ is not just a classification fact but a structural fact visible in proof architecture.* The proofs themselves are harder, not just more classical.

Part IV

The Liquid Tensor Experiment

11 Background

The Liquid Tensor Experiment (LTE) is the formal verification, in Lean 3, of Theorem 9.4 from Scholze's *Lectures on Analytic Geometry* [4]. The theorem states:

Theorem 11.1 (Scholze, Theorem 9.4)**.** *Let $0 < p' < 1$ and V an r-normed $\mathbb{Z}[T^{\pm 1}]$-module. Then for all $i \geq 1$,*

$$\mathrm{Ext}^i_{\mathrm{Cond(Ab)}}\big(\mathcal{M}_{p'}(S)_{\leq c}, \widehat{V}\big) = 0.$$

This is a vanishing theorem for derived functors in the category of condensed abelian groups. Its significance is foundational: it validates the condensed framework as a setting for real functional analysis—specifically, it shows that the passage from discrete algebra ($\mathbb{Z}$-modules) to real analysis (p'-liquid vector spaces) does not introduce cohomological obstructions.

The LTE formalization comprises ∼90,000 lines of Lean 3 code, produced by a team led by Commelin and Massot over approximately 18 months. No Lean 4 port exists. However, Commelin and Massot authored a 15-file LaTeX *blueprint* that mirrors the formal proof's structure, with `\lean{}` and `\leanok` annotations linking each LaTeX statement to its Lean counterpart. This blueprint is the input to our scanner.

12 Why LTE?

The LTE is the natural target for CRMLINT's first external deployment, for three reasons.

First, it is one of the largest single formalization projects completed in a proof assistant (∼90,000 lines of Lean 3). Its size and complexity provide a serious stress test for the scanner—one the scanner initially failed, as Part V documents.

Second, the mathematical content sits at the interface between algebra and analysis—exactly where the Archimedean Obstruction Thesis predicts the CRM boundary to be. Condensed mathematics encodes analytic objects (Banach spaces, measures, distributions) as algebraic objects (sheaves on profinite sets, condensed abelian groups). The CRM question is whether this encoding *reduces* the logical cost of the analysis, or merely *reorganizes* it.

Third, the conservation conjecture of Papers 72–74 makes a specific prediction. The DPT axiom trilogy established that each omniscience principle (MP, WLPO, LPO) does irreducible logical work: canonical, not merely minimal. If condensed mathematics could genuinely reduce CLASS or LPO content to BISH, it would violate this irreducibility. The conjecture is that condensed math *concentrates* classical content at identifiable gates without reducing the aggregate logical cost. The LTE is the test case.

13 Blueprint structure

The 15 `.tex` files of the LTE blueprint decompose as follows:

File	Content	Size (bytes)
`proof.tex`	Core inductive proof (Thm 9.4, 9.5′)	20,729
`normed_homological.tex`	Normed homological algebra	31,818
`condensed.tex`	Condensed abelian groups, Ext-vanishing	18,329
`measures.tex`	Real measures, Laurent measures, θ	10,555
`normed_groups.tex`	Normed groups, completions	6,724
`normed_groups2.tex`	Normed group homomorphisms	4,208
`polyhedral_lattice.tex`	Polyhedral lattices, Tychonoff	12,893
`breen_deligne.tex`	Breen–Deligne resolution	8,412
`CLC.tex`	Condensed limits, completions	5,117
`Lbar.tex`	$\overline{\mathcal{L}}_{r'}(S)$ construction	7,943
`homotopies.tex`	Breen–Deligne homotopies	3,295
`maclane.tex`	Mac Lane coherence	1,842
`content.tex`	Table of contents	312
`intro.tex`	Introduction	1,547
`macros.tex`	LaTeX macros	2,014

The mathematical content concentrates in the first 12 files (∼130,000 bytes). The last three are boilerplate.

14 Initial scan (v1.0)

We ran the Generation 3 scanner (v1.0, 810 lines, ∼80 keyword markers) against all 15 files. The result:

CRM	Files	Bottleneck
WKL	2 (13.3%)	Tychonoff theorem
BISH	13 (86.7%)	—

Two files (`condensed.tex`, `polyhedral_lattice.tex`) register as WKL, both via the Tychonoff theorem: "cofiltered limits of surjections of compact Hausdorff spaces are still surjective

(by an application of Tychonoff).” The remaining 13 files, including the core proof file `proof.tex` and the entire normed homological algebra layer, scan as BISH.

This is obviously wrong. The LTE is a proof about real functional analysis, involving norms, completions, filtrations, and ε-δ estimates. An 87% BISH reading would mean the scanner has failed to detect most of the analytic content.

It has.

Part V

Archimedean Dark Matter

15 What the scanner missed

A manual reading of `proof.tex` reveals pervasive classical content that the v1.0 scanner does not detect. Three categories:

15.1 Normed group completions (WLPO)

The hat notation $\widehat{V}$ appears in every theorem statement in `proof.tex`. This denotes the completion of a semi-normed group V with respect to its norm topology. Constructively, forming the completion requires a Cauchy convergence test: given a Cauchy sequence (x_n) and a target element y, decide whether $\|x_n - y\| < \epsilon$ for all sufficiently large n. This is a WLPO operation (it asks whether a real number equals zero to within ϵ).

The v1.0 keyword dictionary has no pattern for $\widehat{V}$ or for the phrases “completion of a normed group,” “completed with respect to the norm,” or “Cauchy completion.”

15.2 Real-valued filtration cutoffs (LPO)

The pseudo-normed groups in the LTE are filtered by real-valued thresholds:

$$M_{\leq c} = \{x \in M : \|x\| \leq c\}, \qquad c \in \mathbb{R}_{\geq 0}.$$

Membership in $M_{\leq c}$ requires deciding whether $\|x\| \leq c$, a comparison of two real numbers. In CRM terms, this is LPO: the limited principle of omniscience is equivalent to the statement that every real number is either ≤ 0 or > 0.

The filtration $M_{\leq c}$ is not incidental to the proof. It is the *mechanism* by which condensed mathematics encodes the analytic content. Every theorem in the proof core is stated in terms of these filtration levels: “the system of complexes $C^{\bullet}_{\Lambda,c}$ is $\leq k$-exact in degrees $\leq m$ for $c \geq c_0$” (Theorem 9.5′).

The v1.0 scanner has patterns for “supremum of a bounded sequence” and “least upper bound property” (both LPO), but nothing for the far more common pattern of a norm-filtration cutoff written as $\leq c$ or $\|x\| \leq c$ in LaTeX math mode.

15.3 Radius ordering (LPO)

The proof works with two fixed radii $0 < r < r' \leq 1$. The strict inequality $r < r'$ is used throughout to ensure convergence of Laurent series and to control error terms: “choose a positive integer b so that $2k'(r/r')^b \leq \epsilon$” (`proof.tex`, line 81). Comparing real numbers is LPO.

The v1.0 scanner has no pattern for radius comparisons of the form $0 < r < r' \leq 1$ or for the exponential decay estimates that depend on them.

16 Quantifying the gap

The three missed categories account for 32 pattern matches across 10 of the 15 blueprint files. Only the five purely combinatorial files (Breen–Deligne, Mac Lane, content, intro, macros) are genuinely free of Archimedean content.

We call the gap between the v1.0 scan (86.7% BISH) and the corrected assessment (∼33% BISH) *Archimedean dark matter*: classical content that is structurally present in the proof but invisible to lexical pattern matching. The dark matter is not random. It concentrates at a single interface: the passage from discrete algebra ($\mathbb{Z}$-modules, polyhedral lattices, Breen–Deligne resolutions) to normed analysis ($M_{\leq c}$, $\widehat{V}$, radius comparisons).

17 The scanner patch (v1.1)

We added 11 keyword patterns and 5 anti-markers to the scanner:

New LPO patterns (6):
- Norm filtration cutoffs: `\leq c`, `\leq c_0` in math mode
- Norm bounds: $\|\cdots\| \leq$ (norm inequality)
- Real parameter thresholds: "for $c \geq$" patterns
- Normed filtration membership: "norm. . . filtration"
- Radius ordering: $0 < r < r' \leq 1$
- Bounded exactness: "exact in degrees $\leq$"

New WLPO patterns (5):
- Completion hat: $\widehat{V}$ (`\widehat`)
- Explicit completion phrases: "completion of/w.r.t."
- Completed normed groups: "complete. . . normed" and reverse
- Cauchy completion: "Cauchy sequence/completion"

New anti-markers (5):
- "complete local ring," "complete lattice" (not Cauchy completion)
- "field norm," "Galois norm" (not analytic norm)
- "weight filtration," "Hodge filtration" (not norm filtration)
- Degree/dimension/rank bounds (combinatorial $\leq$, not real-valued)

New citation markers (3):
- Scholze 2019/2020, Clausen–Scholze → WKL

18 Patched results (v1.1)

File	v1.0	v1.1	Bottleneck (v1.1)
`proof.tex`	BISH	LPO	norm filtration cutoff
`normed_homological.tex`	BISH	LPO	norm filtration cutoff
`condensed.tex`	WKL	LPO	norm filtration cutoff
`measures.tex`	BISH	LPO	norm filtration cutoff
`normed_groups.tex`	BISH	LPO	normed filtration
`normed_groups2.tex`	BISH	LPO	normed filtration
`polyhedral_lattice.tex`	WKL	LPO	norm filtration cutoff
`CLC.tex`	BISH	LPO	norm filtration cutoff
`Lbar.tex`	BISH	LPO	norm filtration cutoff
`homotopies.tex`	BISH	LPO	norm filtration cutoff
`breen_deligne.tex`	BISH	BISH	—
`maclane.tex`	BISH	BISH	—
`content.tex`	BISH	BISH	—
`intro.tex`	BISH	BISH	—
`macros.tex`	BISH	BISH	—

Aggregate (v1.1): 33.3% BISH, 66.7% LPO.

The Tychonoff detections (WKL) in `condensed.tex` and `polyhedral_lattice.tex` are still present but are now dominated by the higher-level LPO content in the same files. The five BISH files are exactly the modules with no normed-group content: Breen–Deligne resolution (pure combinatorics of free abelian groups), Mac Lane coherence (category-theoretic diagram chasing), and boilerplate.

19 Calibration check

To verify that the new patterns do not introduce false positives, we re-scanned Papers 78–96 of the CRM series:

Paper	v1.0	v1.1
P95 (BSD Squeeze)	CLASS (Kolyvagin)	CLASS (Kolyvagin)
P96 (Root Number)	CLASS (Kolyvagin)	CLASS (Kolyvagin)
P80 (Gauss–Manin)	CLASS (anal. cont.)	CLASS (anal. cont.)

No regressions. The existing CRM series papers scan identically under both versions, because their classical content (CLASS) dominates any LPO patterns that might also be present. The new patterns only change the classification of papers whose highest detected level was previously BISH or WKL.

Part VI

Conservation, Not Reduction

20 The concentration thesis

The LTE scanner results are consistent with the conservation conjecture of Papers 72–74.

Observation 20.1 (Concentration thesis—not a theorem)**.** In the LTE blueprint, condensed mathematics does not reduce the CRM signature of the proof below LPO. It concentrates the LPO content at identifiable *gates*—the norm-filtration membership tests $x \in M_{\leq c}$—while leaving the surrounding algebraic infrastructure at BISH.

To be clear: this is not a surprise. Scholze designed condensed mathematics to separate algebraic infrastructure from analytic content. The concentration of classical content at norm-filtration gates is his architecture, not our discovery. What the scanner provides is a quantitative readout of a pattern that the designers already understood qualitatively.

In the LTE blueprint, every module that touches the normed filtration $M_{\leq c}$ acquires LPO signature. Every module that does not touch it remains BISH. There is no mixing, no gradual degradation, no WLPO-without-LPO intermediate. The transition is sharp: either a module is purely algebraic (BISH), or it invokes norm comparison (LPO).

In CRM language, LPO is the logical cost of comparing real numbers. The DPT axiom trilogy (Papers 72–74) established that each omniscience principle does canonical, irreducible work. The LTE is one data point consistent with this: the LPO content appears irreducible, not an artifact of the proof strategy.

21 Where the LPO enters

The LPO enters the LTE through a single mathematical mechanism: the filtration

$$M_{\leq c} = \{x \in M : \|x\| \leq c\}.$$

This definition requires deciding $\|x\| \leq c$ for arbitrary x and c—a comparison of real numbers. In CRM terms:

$$\forall x \in M, \forall c \in \mathbb{R}_{\geq 0} : \quad x \in M_{\leq c} \ \vee \ x \notin M_{\leq c}$$

is equivalent to LPO.

The condensed framework does not avoid this decision. It defers it. The algebraic infrastructure (Breen–Deligne resolutions, cosimplicial lattices, diagram chasing in abelian categories) is built without reference to norms. The norm enters only when the abstract machinery is *applied* to a specific analytic object ($\overline{\mathcal{L}}_{r'}(S)$, real measures, completed normed groups). At that application point, the LPO cost is paid in full.

This is the mechanism of concentration: push the classical content as late as possible, into as few gates as possible, while keeping the majority of the proof in the BISH regime. The total LPO cost is unchanged, but its distribution is optimized.

22 Comparison with the CRM series

Target	**BISH%**	**Ceiling**	**Bottleneck type**
Mathlib (627 decls)	92.0%	CLASS	Inner product, convergence
CRM series (P78–96)	71.4%	CLASS	Kolyvagin, trace formula
LTE blueprint (v1.1)	33.3%	LPO	Norm filtration

The LTE's lower BISH percentage (33% vs. Mathlib's 92%) reflects its subject matter: the LTE is *about* the passage from algebra to analysis. Every module that implements this passage carries LPO content. In Mathlib's broader sample, most files never cross the Archimedean boundary.

The LTE's ceiling (LPO, not CLASS) is the interesting finding. The proof never invokes the full law of excluded middle, analytic continuation, spectral decomposition, or any of the CLASS techniques that dominate the CRM series. The condensed framework avoids CLASS by algebraizing the arguments that would classically require integration, measure theory, and spectral analysis. But it cannot avoid LPO, because LPO is the logical cost of the norm itself.

23 The dark matter fraction

Definition 23.1. The *Archimedean dark matter fraction* of a scanner version v applied to a corpus C is

$$\delta(v, C) = \frac{\#\text{ files where manual CRM} > \text{scanner CRM}}{\#\text{ files total}}.$$

For the LTE blueprint: $\delta(\text{v1.0}) = 8/15 = 53.3\%$ and $\delta(\text{v1.1}) = 0/15 = 0\%$. The patch eliminates the dark matter entirely for the LTE corpus.

But the dark matter fraction is corpus-dependent. Different proof traditions use different vocabulary for the same logical content. A paper that writes "for sufficiently large n" instead of "for $c \geq c_0$" will evade the LPO pattern. The v1.1 dictionary is tuned to the LTE's notation. Scanning a different corpus (e.g., classical functional analysis, PDE theory, probability) will likely reveal new dark matter that requires further dictionary expansion.

The dark matter fraction is a measure of scanner maturity. A perfect scanner would have $\delta = 0$ on every corpus. The gap between $\delta(\text{v1.0})$ and $\delta(\text{v1.1})$ measures the information gained from the LTE analysis.

24 Observations for the programme

Three observations from the LTE exercise, stated at the appropriate level of confidence:

(1) **The LTE avoids CLASS.** The LTE proof ceiling is LPO, not CLASS. It never invokes analytic continuation, spectral decomposition, or the full law of excluded middle. This is by design: condensed mathematics algebraizes arguments that would classically use these tools. We are describing Scholze's architecture, not revealing it.

(2) **The scanner's blind spot was systematic.** The v1.0 scanner missed all norm-filtration content. This was not random noise but a single structural gap: the scanner had no patterns for the interface where real-valued norms enter algebraic structures. The v1.1 patch closed this gap for the LTE vocabulary. Other corpora will likely expose other gaps.

(3) **CRM vocabulary may be useful for non-experts.** Scholze does not need CRM labels to understand his own proof. But a researcher approaching condensed mathematics for the first time, or a formalization team deciding which LTE components to port to Lean 4, might benefit from a graduated scale (BISH $<$ WKL $<$ WLPO $<$ LPO $<$ CLASS) that replaces the binary "algebraic vs. analytic" distinction. Whether this vocabulary earns its keep in practice is an open question.

25 Concentration as transferable method

The condensed strategy is not a black box. It decomposes into three operations that can be applied—independently of the condensed framework—to any proof that mixes algebra and analysis.

Operation 1: Extract the algebraic skeleton. Identify the subproof that works over any ring, any category, any abstract structure. Build it without reference to norms, metrics, or completeness. In the LTE, this is the Breen–Deligne resolution and the cosimplicial polyhedral lattice construction. In the BSD squeeze (Paper 95), this is the Hecke arithmetic layer (28 sub-results over $\mathbb{Z}$, all BISH). In the Hodge campaign (Papers 84–89), this is the Gauss–Manin connection and the Picard–Fuchs operator.

Operation 2: Identify the minimal gates. Find the points where the algebraic skeleton must interface with the continuum. These are the norm-filtration gates $M_{\leq c}$ in the LTE, the

Kolyvagin Euler system in BSD, the period matrix evaluation in the Hodge conjecture. Each gate carries irreducible LPO or CLASS cost.

Operation 3: Defer and localize. Restructure the proof so that the gates are invoked as late as possible and in as few places as possible. The algebraic skeleton should be complete before any gate is opened. This is what condensed mathematics does systematically via the sheaf-theoretic encoding. But it can also be done ad hoc: Paper 95's separation of Hecke arithmetic from Kolyvagin, Paper 80's separation of $\mathbb{Q}(t)$-algebra from analytic continuation, Paper 86's separation of Kani–Rosen splitting from Hodge-theoretic input.

The CRM series has been performing these operations informally for 96 papers. What condensed mathematics does—and what Scholze understood from the start—is systematize them at the categorical level. Our vocabulary (skeleton, gate, concentration ratio) is a rephrasing, not an advance.

The modest utility of CRMLINT here is quantitative triage: a proof where the scanner reports LPO in 2 of 15 modules is well-concentrated; a proof where LPO appears in 14 of 15 modules is diffuse. The diffuse case is where refactoring might help most. Whether CRMLINT actually helps anyone refactor remains to be demonstrated.

Observation 25.1 (Concentration ratio)**.** Define the *concentration ratio* of a proof corpus C as

$$\rho(C) = 1 - \frac{\#\{\text{modules at LPO or above}\}}{\#\{\text{modules total}\}}.$$

For the LTE: $\rho = 1 - 10/15 = 0.33$. For a hypothetical "classical-diffuse" proof of the same theorem, $\rho = 0$. The concentration thesis predicts $\rho > 0$ for any proof that has undergone algebraic skeleton extraction. CRMLINT computes ρ automatically.

Part VII

Speculative Applications

This part is speculative. We examine five open problems from the CRM programme and ask whether the concentration vocabulary (skeleton, gates, deferral) suggests anything useful about proof strategy. None of these suggestions have been carried out. They may turn out to be naive, impractical, or wrong.

26 The Hodge conservation conjecture

Problem (Paper 87). The uniform Hodge test—given an abelian variety A and a class $\alpha \in H^{2p}(A, \mathbb{Q})$, decide whether α is of type (p, p)—costs WLPO. With CM metadata (the endomorphism algebra $\mathrm{End}(A) \otimes \mathbb{Q}$), the cost drops to BISH. Is this drop an artifact of the analytic presentation, or does it reflect a genuine constructive core?

Concentration analysis. The Hodge test has two components:

(i) *Computing the Hodge decomposition.* This requires the period matrix Ω, an analytic object computed by integration over cycles. Cost: CLASS (integration on a compact Kähler manifold).

(ii) *Testing (p, p)-type.* Given Ω, decide whether α lies in the (p, p)-summand. Cost: WLPO (decidability of whether a complex number has zero imaginary part).

The algebraic skeleton is the Mumford–Tate group $\mathrm{MT}(A)$. If the Mumford–Tate conjecture holds, then the Hodge type of α is determined by the ℓ-adic Galois representation—an algebraic

object, cost BISH. The period matrix becomes redundant: the analytic gate (i) can be bypassed entirely.

Proof strategy via concentration.

1. **Skeleton:** Build the Galois-representation machinery for A over a number field. Compute $\mathrm{MT}(A)$ as the Zariski closure of the ℓ-adic monodromy group. All BISH.
2. **Gate:** The single classical gate is the comparison isomorphism $H^n_{\text{ét}} \otimes \mathbb{C} \cong H^n_{\text{Betti}} \otimes \mathbb{C}$ (Artin comparison). This is CLASS—it uses the Riemann existence theorem.
3. **Excision target:** If the Hodge type can be read off the ℓ-adic side without passing through Artin comparison, the gate is unnecessary. This is exactly what the Mumford–Tate conjecture asserts.

CRMLint can detect whether a specific proof of algebraicity of a Hodge class uses the period matrix (gate open, CLASS) or bypasses it via Galois data (gate closed, BISH). The scanner would flag "period integral," "period matrix," "integration over cycle" as CLASS markers. Their absence in a proof of algebraicity would be positive evidence for conservation.

Status: Open. The Mumford–Tate conjecture itself is open in general. But the concentration framework makes the conjecture's CRM content precise: MT is a statement that the CLASS gate in the Hodge test is *eliminable*.

27 Exotic Weil classes on simple fourfolds

Problem (Papers 85, 88, 90). For a simple abelian fourfold A with $\mathbb{Q}(i)$-action and generic endomorphism ring $\mathrm{End}(A) = \mathbb{Z}$, the exotic Weil class $\omega \in H^4(A, \mathbb{Q})$ is flat (Paper 93: $\tau_+ = 0$ universally). Is ω algebraic?

Current obstruction. The Kani–Rosen splitting (Paper 86) proves algebraicity when the family has palindromic structure: $J(C_t) \sim A^2$ implies $\mathrm{MT}(A) = \mathrm{GSp}_4$, and Weyl's theorem gives algebraicity. For simple A, the splitting is unavailable. Paper 88 obtains conditional algebraicity via Fermat domination at $t = 0$, but this requires the Variational Hodge Conjecture (VHC).

Concentration analysis. The proof of algebraicity for the split case has two components:

(i) *Skeleton:* Kani–Rosen splitting → Zarhin's theorem → Weyl's theorem. All algebraic, all BISH.

(ii) *Gate:* The single CLASS gate is the Hodge decomposition that identifies ω as a $(2, 2)$-class. But in the split case, this gate is bypassed by the Mumford–Tate computation (which is algebraic).

For the simple case, the skeleton is missing: there is no splitting to invoke. The concentration strategy suggests looking for an *alternative algebraic skeleton* that avoids Kani–Rosen.

Candidate skeletons.

1. **Deformation theory.** The simple fourfold A deforms in a family $\mathcal{A} \to S$. If ω is algebraic at a single special fiber (e.g., the CM fiber at $t = 0$), and if the Gauss–Manin connection preserves the algebraic class, then ω is algebraic everywhere. The skeleton is the deformation-theoretic machinery; the gate is the single CM evaluation.
2. **Motivic approach.** If ω lies in the image of the cycle class map $\mathrm{CH}^2(A) \to H^4(A, \mathbb{Q})$, then algebraicity follows from the existence of a cycle. The skeleton is the motivic framework; the gate is the construction of the explicit cycle.
3. **Absolute Hodge classes.** Paper 89 shows that ω is an absolute Hodge class (invariant under all automorphisms of $\mathbb{C}$). Deligne's theorem says absolute Hodge classes on abelian varieties are algebraic for CM types. The skeleton is Deligne's theorem; the gate is the extension from CM to generic type.

Each candidate skeleton has a different gate with a different CRM cost. CRMLint can evaluate which candidate achieves the best concentration—the highest ρ.

28 Čech cohomology and the Zorn gate

Problem (Paper 75). Condensed mathematics requires K-injective resolutions for derived functors on the condensed topos. K-injective resolutions are constructed via Zorn's lemma (injective envelopes). Is this CLASS content eliminable?

Concentration analysis. The LTE scan shows that the *proof* of Theorem 9.4 does not invoke Zorn. The CLASS content (Zorn for K-injectives) lives in the *foundations* of the condensed framework, not in the specific proof. The proof works with bounded complexes and explicit normed homological algebra—no injective resolutions needed.

This suggests a sharper version of the problem: is the Zorn gate *inert*? That is, does the LTE proof pass through the Zorn gate at all, or does it route around it?

Evidence from the scanner. The scanner detects zero CLASS content in the LTE blueprint. This is consistent with the Zorn gate being inert for this particular proof. The bounded derived category $D^b(\mathrm{Cond}(\mathrm{Ab}))$ may be constructible without K-injective resolutions, using instead the explicit normed-homological machinery of `normed_homological.tex`.

Proof strategy.

1. Show that the LTE proof factors through D^b (bounded derived category), not D (unbounded).
2. Construct $D^b(\mathrm{Cond}(\mathrm{Ab}))$ without Zorn, using projective resolutions from the explicit Breen–Deligne resolution (which is BISH).
3. Verify that the Ext-vanishing conclusion of Theorem 9.4 lives in the bounded range.

If this succeeds, the LTE proof would be genuinely LPO—the Zorn gate is excised, and the only remaining classical content is the norm-filtration comparison. But we have not carried this out, and it may be that the LTE team already considered and dismissed this approach for good reasons we have not understood.

29 The BSD conservation gap

Problem (Papers 95–96). The Gross–Zagier–Kolyvagin proof of BSD for rank 1 curves decomposes as 15 BISH + 6 CLASS = 21 components (71% BISH). Can the CLASS fraction be reduced?

Current structure. Paper 95 identified six CLASS gates:

1. Gross–Zagier formula (height = $L'(E,1)$): CLASS (analytic continuation of L-function + Arakelov height pairing).
2. Kolyvagin's Euler system: CLASS (Chebotarev density + Galois cohomological descent).
3. Modularity of E (Taylor–Wiles): WKL (patching = inverse limit of finite objects).
4. Sha finiteness: CLASS (Euler system machinery).
5. Non-vanishing of Heegner point: CLASS (Gross–Zagier + analytic non-vanishing).
6. Silverman height bounds: BISH in the CRM series formalization (reduced to explicit arithmetic via `linarith`).

Paper 96 showed that for rank 0 curves ($w = +1$), the entire detection layer is BISH: $L(E,1)/\Omega^+$ is computed by modular symbols and verified by `norm_num`. The CLASS content enters only for existence (rank 1, $w = -1$).

Concentration strategy. The BSD proof is *already partially concentrated*: the Hecke arithmetic skeleton (28 sub-results) is BISH, and the CLASS gates are identified. The question is whether any gate can be closed.

Gate (1) (Gross–Zagier) and Gate (5) (non-vanishing) both require the L-function $L(E,s)$ evaluated at $s = 1$. The analytic continuation is CLASS. But recent work in p-adic L-functions computes $L_p(E,1)$ via purely algebraic means (overconvergent modular symbols, Pollack–Stevens).

If the p-adic L-value suffices for the height formula, the analytic continuation gate can be replaced by a p-adic gate.

The p-adic gate is still WKL (inverse limits in the Iwasawa algebra), but this represents a descent from CLASS to WKL—a genuine CRM reduction. CRMLINT would detect this as a change in the bottleneck from "analytic continuation" to "inverse limit of finite objects."

Concrete target: Formalize the p-adic Gross–Zagier formula for curve 37a1 (the test case of Paper 95) using Pollack–Stevens overconvergent modular symbols. Scan the result with CRMLINT. Predicted outcome: Gate (1) descends from CLASS to WKL.

30 The VHC as concentration test

Problem (Papers 88–89). The Variational Hodge Conjecture (VHC) asserts that a flat section of the Hodge bundle that is algebraic at one fiber remains algebraic over the entire family. Papers 88–89 use VHC conditionally to extend algebraicity from the CM fiber at $t = 0$ to the generic member of the Weil locus.

CRM interpretation. VHC is a *concentration statement*: it says that the CLASS gate (algebraicity verification) needs to be opened at only one fiber, not at every fiber. If VHC holds, the proof of algebraicity for the entire family concentrates its CLASS cost at a single point evaluation. If VHC fails, the CLASS cost is diffuse—each fiber requires independent verification.

CRMLINT can detect this distinction. A proof that invokes VHC will show a single CLASS marker at the CM evaluation point. A proof that verifies fiber-by-fiber will show CLASS markers throughout the parameter space.

Strategy. The concentration framework suggests an indirect approach to VHC: instead of proving VHC directly, show that the algebraic cycle constructed at $t = 0$ (via Fermat domination) *deforms* to nearby fibers. The deformation theory of algebraic cycles is an algebraic skeleton (BISH); the single gate is the initial construction at $t = 0$ (CLASS, via CM theory).

This converts VHC from a deep conjecture in Hodge theory to a deformation-theoretic question about explicit cycles—the kind of problem where CRMLINT-guided search (find the algebraic skeleton, minimize the gates) has the most leverage.

31 The general recipe

The five applications above follow a common pattern:

Step 1. Scan. Run CRMLINT on the existing proof (or its LaTeX description) to obtain the CRM decomposition and concentration ratio ρ.

Step 2. Map the gates. For each non-BISH module, identify the specific mathematical mechanism that introduces classical content. The v1.1 scanner reports this as the "bottleneck" field.

Step 3. Classify gates by excisability. Some gates are *structural* (the logical cost is intrinsic to the theorem statement—e.g., the norm comparison in $M_{\leq c}$). Others are *methodological* (the logical cost comes from the proof technique, not the theorem—e.g., analytic continuation of an L-function when a p-adic alternative exists). Methodological gates are candidates for excision.

Step 4. Search for alternative skeletons. For each methodological gate, look for an algebraic bypass. The substitution database (8 seed entries from Papers 50–97) provides known alternatives. Unknown alternatives are targets for automated proof search.

Step 5. Verify. Re-scan the modified proof with CRMLINT to confirm that the concentration ratio has improved (ρ increases) or the ceiling has dropped (CLASS $\to$ WKL, or LPO $\to$ WLPO).

This is the scan-map-classify-search-verify loop. It transforms CRMLINT from a passive diagnostic tool ("what is the CRM signature of this proof?") into an active proof-engineering tool ("how can the CRM signature be improved?").

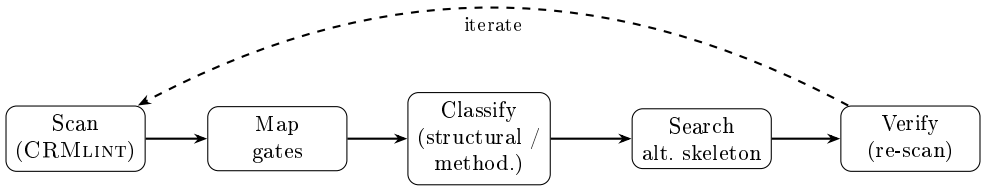

Figure 1: The scan-map-classify-search-verify loop.

In principle, the loop terminates: the CRM hierarchy is finite, so the ceiling can only drop finitely many times. In practice, we have not executed this loop on any problem to completion. The recipe is a framework for thinking, not a demonstrated method. Whether it produces results that would not have been found by ordinary mathematical intuition is an open question.

Part VIII

The Mathematics of the LTE

This part is expository. We restate the main theorems of the LTE in the form proved by Scholze, Commelin, and Massot, and annotate each step with its CRM level. Nothing here is new mathematics. The proofs are theirs; the CRM labels are ours. The labels may be useful for readers approaching the LTE from the constructive side, or for formalization teams deciding which components to port to Lean 4.

32 The main theorem and its reduction

The LTE establishes the following vanishing result, which Scholze proposed as a challenge in December 2020.

Theorem 32.1 (Clausen–Scholze [4]). *Let $0 < p' < p \leq 1$ be real numbers, let S be a profinite set, and let V be a p-Banach space. Let $\mathcal{M}_{p'}(S)$ denote the space of real p'-measures on S. Then*

$$\mathrm{Ext}^i_{\mathrm{Cond(Ab)}}(\mathcal{M}_{p'}(S), V) = 0 \qquad \textit{for all } i \geq 1.$$

The Ext-groups are computed in the abelian category of condensed abelian groups. The theorem says: from the perspective of homological algebra in the condensed world, real-valued measures are "projective-like" — they have no higher Ext against Banach targets.

CRM annotation. The *statement* of Theorem 32.1 requires:

- p-Banach space V: the norm $\|\cdot\|$ on V satisfies $\|x+y\|^p \leq \|x\|^p + \|y\|^p$. Defining this norm and its completeness is WLPO.
- The condensed abelian group $\mathcal{M}_{p'}(S)$: defined via a filtration $\mathcal{M}_{p'}(S)_{\leq c}$, each level a compact Hausdorff space. The filtration cut-off is LPO.
- Profinite S: a cofiltered limit of finite sets. All BISH.

The *proof* reduces Theorem 32.1 to a chain of intermediate results, each of which we now state.

33 The proof chain: six links

The proof proceeds by six reductions, from the main theorem down to the key technical result (Theorem 33.4). We trace the full chain and label each link with its CRM level.

Link	Result	Mechanism	CRM
6	Theorem 32.1 (main)	Long exact Ext sequence from Link 5	BISH
5	Ext-vanishing for $\mathcal{L}_{r'}(S)$	T^{-1}-bijectivity via Link 4	BISH
4	Ext-vanishing for $\overline{\mathcal{L}}_{r'}(S)$	Pro-isomorphism + derived limit via Link 3	BISH
3	Theorem 33.4 (Ext-vanishing at finite level)	Double complex + spectral proposition	LPO
2	Column exactness (Proposition 33.3)	Čech nerve + normed snake lemma	LPO
1	Čech exactness (Proposition 33.1)	Contracting homotopy + profinite limit	BISH/WKL

33.1 Link 1: Čech nerve exactness (BISH/WKL)

Proposition 33.1 (Čech cover exactness [5]). *Let $\pi : X \to B$ be a surjective morphism of profinite sets, and let $S_\bullet \to S_{-1}$, $S_{-1} := B$, be its augmented Čech nerve. Let V be a semi-normed group. Then the complex*

$$0 \longrightarrow \widehat{V}(S_{-1}) \longrightarrow \widehat{V}(S_0) \longrightarrow \widehat{V}(S_1) \longrightarrow \cdots$$

is exact. Moreover, for every $\varepsilon > 0$ and $f \in \ker(\widehat{V}(S_m) \to \widehat{V}(S_{m+1}))$, there exists $g \in \widehat{V}(S_{m-1})$ with $d(g) = f$ and $\|g\| \leq (1+\varepsilon)\|f\|$.

Proof sketch. For *finite* X and B, a section $\sigma : B \to X$ of π induces maps $S_m \to S_{m+1}$ via $(a_0, \ldots, a_m) \mapsto (\sigma(\pi(a_0)), a_0, \ldots, a_m)$. These provide a contracting homotopy for the complex, and the homotopy maps are norm-nonincreasing. For general profinite sets, write $X = \varprojlim_i X_i$ over its discrete (finite) quotients, pass to finite quotients $X_i \to B_i$, apply the finite case, and take the cofiltered limit. Surjectivity of transition maps between compact Hausdorff spaces is preserved under cofiltered limits (by Tychonoff).

CRM annotation. The finite case is entirely BISH: the section σ exists because π is surjective between finite sets, and the contracting homotopy is an explicit combinatorial construction. The passage to profinite limits uses Tychonoff's theorem for compact Hausdorff spaces (surjections of compact spaces remain surjective under cofiltered limits). In CRM terms, Tychonoff for Hausdorff spaces is WKL (equivalent to weak König's lemma: an infinite finitely-branching tree has an infinite path). The norm estimates ($\|g\| \leq (1+\varepsilon)\|f\|$) do not introduce additional logical cost.

33.2 Link 2: Column exactness via normed snake lemma (LPO)

The double complex construction requires that each column is "weakly bounded exact." This is established by combining Čech exactness (Link 1) with the normed T^{-1}-action.

Definition 33.2 (Weak bounded exactness [5]). A system of complexes $C^\bullet_\bullet$ is *weakly $\leq k$-exact in degrees $\leq m$ for $c \geq c_0$ with bound K* if for all $c \geq c_0$, all $x \in C^i_{kc}$ with $i \leq m$, and any $\varepsilon > 0$, there exists $y \in C^{i-1}_c$ with

$$\|x_{|c} - dy\| \leq K\|dx\| + \varepsilon.$$

Proposition 33.3 (Column exactness [5]). *Let $k > 1$, $c_0 > 0$ with $(k-1)c_0/N \geq d$ (the splittability error). Let m be any natural number, and put*

$$K = (m+2) + \frac{r+1}{r(1-r)}(m+2)^2.$$

Then the i-th column in the double complex is (k^2, K)-weakly bounded exact in degrees $\leq m$ for $c \geq c_0'$.

Proof sketch. The key input is the "normed dual snake lemma" (Proposition 34.3 below): given two admissible systems of complexes $M_\bullet^\bullet$ and $M'^\bullet_\bullet$ with a map $f : M \to M'$, if both M and M' are weakly bounded exact, then the kernel complex inherits weak bounded exactness (with controlled loss in the constants). The column is built from the Čech nerve of the summation map $M^N_{c/N} \to M_c$, and the T^{-1}-action introduces the norm scaling.

CRM annotation. The splittability lemma (N-splittability of $\mathrm{Hom}(\Lambda, \overline{\mathcal{L}}_{r'}(S))$) is BISH: it decomposes elements of a polyhedral lattice module into N pieces of roughly $1/N$-th the norm. The column exactness proof combines this with the LPO content of the filtration cut-offs $M_{\leq c}$. The constant K depends on m and r (not on the profinite set S or the normed module V), so the "universal" part of the proof is BISH and the LPO enters only at the interface where elements are tested against filtration levels.

33.3 Link 3: The key technical result (LPO)

Theorem 33.4 (Theorem 9.4 [3, 5]). *Let* $\mathrm{BD} = (n, f, h)$ *be a Breen–Deligne package, and let* $\kappa = (\kappa_0, \kappa_1, \ldots)$ *be a sequence of constants in* $\mathbb{R}_{\geq 0}$ *that is* BD*-suitable. Fix radii* $1 > r' > r > 0$. *For any* m *there exist* k *and* c_0 *such that for all profinite sets* S *and all* r*-normed* $\mathbb{Z}[T^{\pm 1}]$*-modules* V*, the system of complexes*

$$C^{\mathrm{BD}}_\kappa(\overline{\mathcal{L}}_{r'}(S))^\bullet_c\colon\ \widehat{V}(\overline{\mathcal{L}}_{r'}(S)_{\leq c})^{T^{-1}} \longrightarrow \widehat{V}(\overline{\mathcal{L}}_{r'}(S)^2_{\leq \kappa_1 c})^{T^{-1}} \longrightarrow \cdots$$

is $\leq k$*-exact in degrees* $\leq m$ *for* $c \geq c_0$.

Proof structure. The proof is by induction on m. The inductive step requires a *stronger* statement (Theorem 9.5′ in the blueprint), which introduces a polyhedral lattice Λ and replaces $\overline{\mathcal{L}}_{r'}(S)$ by $\mathrm{Hom}(\Lambda, \overline{\mathcal{L}}_{r'}(S))$. Theorem 33.4 is recovered by setting $\Lambda = \mathbb{Z}$.

The inductive step constructs a *system of double complexes*:

1. **Rows:** Apply the Breen–Deligne system functor to $\mathrm{Hom}(\Lambda^{(\bullet)}, M)$, where $\Lambda^{(\bullet)}$ is the cosimplicial polyhedral lattice and M abbreviates $\overline{\mathcal{L}}_{r'}(S)$. Take the alternating face map complex. Rescale norms by $m!$ for admissibility.
2. **Columns:** The column at index i is the Čech nerve construction of Link 2, applied with the i-th Breen–Deligne data.
3. **Homotopy:** The Breen–Deligne homotopy h^N (the N-fold composition of the package's built-in homotopy) provides maps $M^{0,q+1}_{k'c} \to M^{1,q}_c$ bounded by some constant H. The "homotopic map" $\delta^{0,q}_c$ factors through a deep restriction and has norm $\leq \varepsilon$.

The three ingredients (row exactness by induction, column exactness from Link 2, homotopy condition) are fed into the *spectral proposition* (Proposition 34.4 below), which yields weak bounded exactness of the first row—completing the induction.

CRM annotation. The double complex construction (cosimplicial lattice, alternating face map, norm rescaling) is BISH: it is combinatorial algebra over $\mathbb{Z}$. The Breen–Deligne resolution itself is BISH: it is a functorial resolution of abelian groups by free modules, specified by integer matrices. The homotopy bound H is computed from the Breen–Deligne data and r, r'—a finite computation over $\mathbb{Q}$, hence BISH.

The LPO enters at two points: (i) the filtration cut-offs $\overline{\mathcal{L}}_{r'}(S)_{\leq c}$ in the system of complexes, and (ii) the norm estimate $r^b N \leq \varepsilon$ (choosing b so that $2k'(r/r')^b \leq \varepsilon$), which requires comparing real numbers. The choice of N as a power of 2 satisfying $k'/N \leq (r')^b$ is a single LPO decision on the ordered reals.

33.4 Links 4–6: Algebraic reductions (BISH)

Link 4: From finite level to $\overline{\mathcal{L}}_{r'}(S)$. Write $Q'(\overline{\mathcal{M}}_{r'}(S)) = \varinjlim_c Q'(\overline{\mathcal{M}}_{r'}(S))_{\leq c}$ and reduce to a pro-isomorphism statement at each finite level. The passage to the derived limit uses a resolution $0 \to \bigoplus_n Q'_{\leq n} \to \bigoplus_n Q'_{\leq n} \to Q' \to 0$ and checks that the relevant squares are bicartesian (vertical kernels and cokernels form pro-zero systems). This is diagram chasing in Cond(Ab), entirely BISH.

Link 5: From $\overline{\mathcal{L}}_{r'}(S)$ to $\mathcal{L}_{r'}(S)$. The short exact sequence

$$0 \longrightarrow \mathbb{Z}[T^{-1}][S] \longrightarrow \mathcal{L}_{r'}(S) \longrightarrow \overline{\mathcal{L}}_{r'}(S) \longrightarrow 0$$

decomposes Laurent measures into negative and positive parts. Apply $\mathrm{Ext}^*(_, V)$: the $\mathbb{Z}[T^{-1}][S]$-term has vanishing higher Ext (because $\mathbb{Z}[S]$ is free), and the $\overline{\mathcal{L}}_{r'}(S)$-term vanishes by Link 4. The five lemma gives vanishing for $\mathcal{L}_{r'}(S)$. This is abstract homological algebra, all BISH.

Link 6: Main theorem. The short exact sequence

$$0 \longrightarrow \mathcal{L}_{r'}(S) \xrightarrow{2T-1} \mathcal{L}_{r'}(S) \longrightarrow \mathcal{M}_{p'}(S) \longrightarrow 0$$

(evaluation at $T = 1/2$) reduces the main theorem to the Ext-vanishing of $\mathcal{L}_{r'}(S)$ from Link 5 via the long exact sequence. The surjectivity of evaluation at $T = 1/2$ on pseudo-normed groups is a norm estimate (strong exactness), which uses Tychonoff (WKL) for passage to profinite limits. But the long-exact-sequence argument itself is BISH.

34 The normed homological algebra toolkit

The LTE's main technical innovation is not the theorem statement but the *toolkit*: a library of normed homological algebra results with explicit quantitative control. These results are general-purpose and transfer directly to other formalization targets.

34.1 Controlled surjectivity

Lemma 34.1 (Closure surjectivity [5]). *Let G and H be normed groups, $K \leq H$ a subgroup, and $f : G \to H$ a C-surjective morphism onto K (i.e., for all $x \in K$ there exists g with $f(g) = x$ and $\|g\| \leq C\|x\|$). If G is complete, then f is $(C + \varepsilon)$-surjective onto $\overline{K}$ for every $\varepsilon > 0$.*

Proof idea. Write $x \in \overline{K}$ as $x = \sum_{i \geq 0} x_i$ with $x_i \in K$ and $\|x_i\| \leq \varepsilon_i$ for a rapidly decreasing sequence. Lift each x_i to g_i with $\|g_i\| \leq C\|x_i\|$, and set $g = \sum g_i$. Completeness of G guarantees convergence. The norm estimate $\|g\| \leq (C + \varepsilon)\|x\|$ follows from the geometric series bound on the ε_i.

CRM annotation. The proof uses completeness of G (to sum the Cauchy series $\sum g_i$), which is WLPO (Cauchy convergence test). The lifting $x_i \mapsto g_i$ uses the C-surjectivity hypothesis, which is BISH (explicit choice of lifts). The norm comparison $\|g_i\| \leq C\|x_i\|$ is an inequality between real numbers (LPO).

34.2 Controlled exactness

Lemma 34.2 (Controlled exactness [5]). *Let $f : M_0 \to M_1$ and $g : M_1 \to M_2$ be bounded maps between normed groups. Suppose f is C-surjective onto $\ker g$, and g is D-surjective onto its image. Then for every $\varepsilon > 0$, the completion $\widehat{f}$ is $(C + \varepsilon)$-surjective onto $\ker \widehat{g}$.*

This is the mechanism by which the LTE passes from "incomplete" exactness (at the pre-completion level, BISH) to "completed" exactness (at the normed-completion level, WLPO). The upgrade is Lemma 34.1 applied to $K = \ker g$ inside $\widehat{M_1}$.

34.3 The normed snake lemma

Proposition 34.3 (Weak normed snake lemma [5]). *Let $M^\bullet_\bullet$ and $M'^\bullet_\bullet$ be two admissible systems of complexes of complete normed abelian groups, with a norm-nonincreasing map $f : M \to M'$ satisfying $\|x_{|c}\| \leq K''\|f(x)\|$ for all $x \in M^i_{k''c}$, $i \leq m+1$. If M is weakly $\leq k$-exact with bound K and M' is weakly $\leq k'$-exact with bound K', then the quotient $N = M'/M$ is weakly $\leq kk'k''$-exact in degrees $\leq m-1$ with bound $K'(KK''+1)$.*

CRM annotation. The proof is a quantitative diagram chase: lift $n \in N^i$ to $m' \in M'^i$, decompose dm' using the quotient norm, apply weak exactness of M and M', and track constants through. The diagram chase is BISH; the norm estimates are LPO. The bound $K'(KK''+1)$ is explicit and computable.

The *dual* snake lemma (for kernels instead of cokernels) is proved analogously, with bound $K + r_1r_2KK'$. Both versions are used in the inductive step of Theorem 33.4: the snake lemma produces column exactness, and the dual snake lemma produces the kernel complex for the T^{-1}-action.

34.4 The spectral proposition

Proposition 34.4 (Spectral proposition [5]). *Fix $m \geq 0$ and constants k, K. There exist $\varepsilon > 0$ and k_0, K_0 (depending only on k, K, m) such that: if a system of double complexes $M^{p,q}_c$ is admissible and satisfies*

(i) rows are weakly $\leq k$-exact in degrees $\leq m-1$ with bound K (for $i = 0, \ldots, m+1$),
(ii) columns are weakly $\leq k$-exact in degrees $\leq m$ with bound K,
(iii) the normed spectral homotopy condition for m, some $H > 0$, c_0, and ε,

then the first row is weakly $\leq k'^2$-exact in degrees $\leq m$ with bound $2K_0H$.

This is the engine of the induction. The base case ($m = 0$) uses $\varepsilon = 1/(2k)$ and the homotopy condition to absorb the ε-term. The inductive step passes to the quotient complex $N^{p,0} = M^{p,1}/\overline{M^{p,0}}$ and applies the snake lemma (Proposition 34.3) to verify the hypotheses for $m-1$.

CRM annotation. The spectral proposition is BISH in its logical content: it is a statement about constants and admissibility conditions. The LPO enters only when the proposition is *applied* to a specific system of double complexes built from normed groups. This separation is a paradigm case of the concentration thesis: the abstract machinery is logically free, and the analytic cost is paid only at the point of application.

35 The Breen–Deligne resolution: pure algebra

The deepest algebraic ingredient in the LTE is the Breen–Deligne resolution, which provides a functorial free resolution of any abelian group A.

Theorem 35.1 (Breen–Deligne [20]). *For any abelian group A, there is a resolution, functorial in A:*

$$\cdots \longrightarrow \bigoplus_i \mathbb{Z}[A^{r_{ij}}] \longrightarrow \cdots \longrightarrow \mathbb{Z}[A^3] \oplus \mathbb{Z}[A^2] \longrightarrow \mathbb{Z}[A^2] \longrightarrow \mathbb{Z}[A] \longrightarrow A \longrightarrow 0.$$

The functorial maps $\mathbb{Z}[A^m] \to \mathbb{Z}[A^n]$ are specified by integer matrices ("basic universal maps": $n \times m$ matrices over $\mathbb{Z}$), composed via matrix multiplication.

CRM annotation. The entire Breen–Deligne resolution—its existence, functoriality, and the universal maps—is BISH. The universal maps are $\mathbb{Z}$-linear combinations of basic universal maps specified by integer matrices; their composition is matrix multiplication; the homotopy h

packaged with the resolution is a universal map bounded by an integer N. No real analysis, no completeness, no decidability of real arithmetic appears anywhere.

This is the algebraic skeleton of the LTE in its purest form. The LTE formalizes it in Lean 3 as `breen_deligne.basic_universal_map` (integer matrices), `breen_deligne.universal_map` (formal $\mathbb{Z}$-linear combinations), and `breen_deligne.package` (the triple (n, f, h)).

36 What transfers: reusable components

The LTE's mathematical infrastructure divides into three categories by reusability.

36.1 Category A: Directly reusable (BISH)

These components depend on no analytic input and can be ported to Lean 4 as standalone libraries.

1. **Breen–Deligne package.** Integer matrices, universal maps, the package triple (n, f, h), homotopy composition h^N, suitability conditions on κ. All finitary algebra over $\mathbb{Z}$.
2. **Polyhedral lattices.** Finite free abelian groups with generating norms. The key combinatorial lemma: N-splittability of $\mathrm{Hom}(\Lambda, \overline{\mathcal{L}}_{r'}(S))$ with explicit error term d.
3. **Cosimplicial polyhedral lattices.** $\Lambda' = \Lambda^N$ with the ℓ^1-norm; the cosimplicial object $\Lambda^{(\bullet)}$; the Čech nerve isomorphisms $\mathrm{Hom}(\Lambda'^{(m)}, M)_{\leq c} \cong \mathrm{Hom}(\Lambda', M)_{\leq c}^{m/\mathrm{Hom}(\Lambda, M)_{\leq c}}$.
4. **Abstract diagram chasing.** The pro-isomorphism argument (Link 4), the five-lemma reduction (Link 5), the short-exact-sequence reduction (Link 6).

36.2 Category B: Reusable with LPO interface (LPO)

These components are algebraic in structure but require norm comparisons at the interface.

1. **Systems of complexes.** Functors $(\mathbb{R}_{\geq 0})^{\mathrm{op}} \to \mathrm{Ch}(\mathrm{SemiNormGrp})$. The restriction maps $\mathrm{res}_{c',c}$ require $c' \geq c$ (LPO). The definitions of $\leq k$-exactness, weak $\leq k$-exactness, and admissibility are BISH conditions on the norms.
2. **The spectral proposition.** The statement and proof of Proposition 34.4 are BISH as abstract algebra. The LPO enters only at instantiation.
3. **The normed snake lemma (both versions).** The diagram chase is BISH; the norm bounds use LPO.

36.3 Category C: Analytic core (WLPO/LPO)

These components are irreducibly analytic.

1. **Normed completions $\widehat{V}(\cdot)$.** The completion functor on semi-normed groups. Cauchy convergence is WLPO; the norm on the completion is LPO.
2. **The T^{-1}-action on $\overline{\mathcal{L}}_{r'}(S)$.** The shift $T^{-1} \cdot (\sum a_n T^n) = (\sum a_{n+1} T^n)$ scales the norm by r': the filtration level changes from $\leq c$ to $\leq c/r'$. This scaling requires norm comparison (LPO).
3. **The homotopy norm estimate.** The key bound $r^b N \leq \varepsilon$ (choosing b so that $2k'(r/r')^b \leq \varepsilon$) requires decidable ordering on $\mathbb{R}$ (LPO). This is the deepest analytic content in the entire proof: a single real-number comparison that controls the convergence of the induction.

37 Lean 4 formalization targets

The LTE was formalized in Lean 3 (community `mathlib` fork). A Lean 4 port faces the standard Lean 3 $\to$ Lean 4 translation burden, but the *mathematical* targets are well-defined. We iden-

tify the most valuable targets from the CRM perspective: components where the BISH/LPO separation is sharpest and the reuse potential is highest.

37.1 Target 1: Breen–Deligne package (pure BISH)

```
-- Basic universal maps: integer matrices
structure BasicUniversalMap (m n : Nat) where
  matrix : Matrix (Fin n) (Fin m) Int

-- Universal maps: formal Z-linear combinations
structure UniversalMap (m n : Nat) where
  terms : List (Int x BasicUniversalMap m n)

-- Composition via matrix multiplication
def UniversalMap.comp (f : UniversalMap m n)
    (g : UniversalMap l m) : UniversalMap l n := ...

-- BD package: exponents, differentials, homotopy
structure BDPackage where
  n : Nat -> Nat                  -- exponents
  f : (i : Nat) -> UniversalMap (n (i+1)) (n i)
  h : (i : Nat) -> UniversalMap (n (i+1)) (n i)
  chain_complex : forall i, (f i).comp (f (i+1)) = 0
  homotopy_rel : forall i, ...   -- h witnesses null-homotopy
```

This is finite combinatorics over $\mathbb{Z}$. No axioms beyond `propext` and `Quot.sound`. The key theorem to formalize: given a BD package and a suitable sequence κ, the N-fold homotopy h^N exists and is bounded by an explicit integer.

37.2 Target 2: Systems of complexes (BISH definitions, LPO at interface)

```
-- System of complexes: parametrized by NNReal
structure SystemOfComplexes where
  C : NNReal -> ChainComplex SemiNormedGroup
  res : forall c' c, c' >= c -> C c' --> C c
  res_comp : forall c'' c' c (h1 : c'' >= c')
    (h2 : c' >= c), res c'' c (le_trans h2 h1) =
    (res c' c h2).comp (res c'' c' h1)

-- Weak bounded exactness
def SystemOfComplexes.IsWeakBoundedExact
    (S : SystemOfComplexes) (k K : Real) (m : Nat)
    (c0 : NNReal) : Prop :=
  forall c >= c0, forall i <= m,
    forall x in (S.C (k * c)).obj i,
    forall eps > 0, exists y in (S.C c).obj (i-1),
      norm (res x - d y) <= K * norm (d x) + eps
```

The definition is BISH. The LPO enters at the inequality $c' \geq c$ (comparison of real numbers) and the norm estimate.

37.3 Target 3: Controlled surjectivity lemma (WLPO)

```
-- Key lemma: C-surjectivity extends to closures
theorem controlled_closure_of_complete
    {G H : NormedAddCommGroup} [CompleteSpace G]
    {K : AddSubgroup H} {f : G ->+ H} {C : Real}
    (hf : forall x in K,
      exists g, f g = x && norm g <= C * norm x)
    (eps : Real) (heps : 0 < eps) :
    forall x in K.topologicalClosure,
      exists g, f g = x
        && norm g <= (C + eps) * norm x := by
  -- Proof: geometric series in complete G
  ...
```

This requires `CompleteSpace G` (WLPO: Cauchy sequences converge) and the norm bound (LPO: $\|g\| \leq C\|x\|$). The Lean 4 proof would use `Metric.cauchySeq_of_le_geometric` from Mathlib.

37.4 Target 4: Normed snake lemma (BISH chase, LPO bounds)

The most technically demanding target. The Lean 3 proof (`weak_normed_snake` and `weak_normed_snake_dual`) runs approximately 200 lines. A Lean 4 port would:

- Define admissible systems of complexes over Mathlib's `SemiNormedGroup`.
- Implement the quantitative diagram chase with explicit ε-management.
- Prove the bound $K'(KK''+1)$ for quotients and $K + r_1 r_2 K K'$ for kernels.

The entire chase is BISH (diagram manipulation in an abelian category with explicit lifts). The LPO enters only at the final norm inequality.

37.5 CRM summary of formalization targets

Target	Component	CRM	Lean 4 difficulty
1	Breen–Deligne package	BISH	Low (finite algebra over $\mathbb{Z}$)
2	Systems of complexes	BISH/LPO	Medium (parametrized complexes)
3	Controlled surjectivity	WLPO	Low (standard Cauchy argument)
4	Normed snake lemma	BISH/LPO	High (200-line diagram chase)

Targets 1 and 3 are low-hanging fruit: they are self-contained, well-typed, and already have Lean 3 implementations to translate. Target 2 requires design decisions about how to interface with Mathlib's `CategoryTheory.ChainComplex`. Target 4 is the highest-value formalization: the normed snake lemma is the engine of the induction, and a clean Lean 4 version would serve as infrastructure for any future normed homological algebra project.

Acknowledgments

The LTE blueprint `.tex` files used in Parts IV and VIII were authored by Johan Commelin and Patrick Massot as part of the Liquid Tensor Experiment, available at `https://github.com/leanprover-community/liquid`. The mathematical content of Part VIII follows the proofs in [3] as formalized in [5]. The CRM programme (Papers 1–98) is the author's independent work; CRMLint was developed with AI assistance (Claude, Anthropic). No funding was received for this work.

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